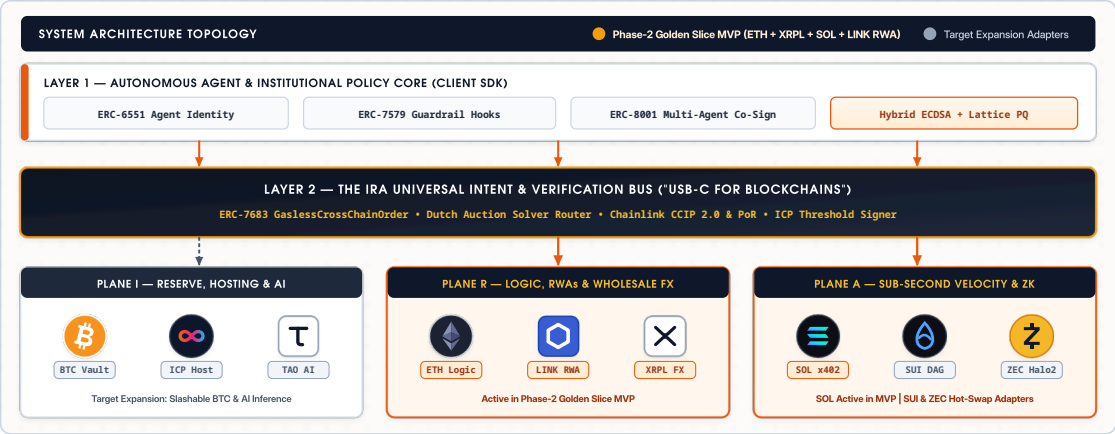
INSTITUTIONAL ARCHITECTURE SPECIFICATION | VERSION 5.1 (FINAL MASTER SPEC) | OCTOBER 2026

THE IRA PROTOCOL

An Anti-Fragile, Intent-Routed Coordination Layer Uniting Pluggable Reserve, Logic, Real-World Asset (RWA) Tokenization, Sub-Second Velocity, and Zero-Knowledge Privacy Adapters for the Autonomous Economy



System Architecture Topology: The IRA Universal Intent Bus separates the **Phase-2 Golden Slice MVP** (Ethereum + XRPL + Solana x402 + ERC-3643/4626 RWA Vaults, highlighted in warm amber-orange) from the full **Target Expansion Topology** (Bitcoin Babylon, ICP, Bittensor, Sui Mysticeti, and Zcash Halo 2).

Abstract & Phased Architectural Scope

Following the conciseness standard of Satoshi Nakamoto's 9-page Bitcoin whitepaper, this specification defines **The IRA Protocol (Intent-Routed Autonomous Protocol)**. Monolithic Layer-1 blockchains cannot simultaneously optimize for 15-year monetary immutability, Turing-complete enterprise compliance, yield-bearing Real-World Asset (RWA) custody, sub-second AI micro-payment velocity, and zero-knowledge commercial privacy. Meanwhile, traditional lock-and-mint bridges expose capital to multi-billion-dollar custodial hacks, and hard-coding multiple blockchains together risks creating an unmaintainable system vulnerable to single-chain outages or censorship.

The IRA Protocol resolves these bottlenecks via a **Lean Universal Intent Core paired with Hot-Swappable Network Adapters ("USB-C for Blockchains")**. The IRA Core standardizes only modular AI agent smart accounts (ERC-6551 / ERC-7579), single-chain multi-agent coordination (ERC-8001), compliant tokenized RWA treasury vaults (ERC-3643 / ERC-4626), hybrid post-quantum authorization (following the emerging EIP-8051 / EIP-8355 Draft precompile proposals), and cross-chain Dutch-auction intents (ERC-7683). To prevent scope creep, the protocol executes in two distinct scopes: a focused **Phase-2 Golden Slice MVP** (Ethereum + XRPL + Solana + 1 RWA Vault) that expands modularly into the full **Target Architecture** across three isolated adapter planes (**I-R-A**):

IRA Adapter Plane	MVP vs. Target Adapters	Isolated Functional Mandate & Failover Guarantee
I — Immutable Reserve & Compute	*Target Expansion:* Bitcoin (Babylon), ICP & Bittensor *Fallback: Native ETH / Akash*	Self-custodial slashable BTC solver bonds via Extractable One-Time Signatures (EOTS); bridge-free threshold signing via ICP Chain Fusion; Yuma-validated AI inference.
R — Regulated Logic, RWAs & Wholesale FX	**Active in MVP:** Ethereum, XRP Ledger, ERC-3643 & XRPL MPT *Fallback: Arbitrum / Base / CCTP*	Modular agent guardrails (ERC-7579), yield-bearing Tokenized Real-World Assets (T-Bills, real estate, credit verified by Chainlink PoR), and 3–5s wholesale FX clearing via XRPL ODL.
A — Autonomous Velocity & ZK Shield	**Active in MVP:** Solana (x402) *Target / Hot-Swap:* Sui Mysticeti & Zcash (Halo 2)	400ms AI micro-payments via HTTP 402 (x402) on Solana with automatic sub-100ms failover to Sui (~350ms Mysticeti DAG); commercial confidentiality with viewing keys via Zcash Orchard.

Architecture Pattern:
Lean Intent Core + Hot-Swap Adapters

Core Standards:
ERC-7683, ERC-7579, ERC-3643, x402, EIP-8051

Prior-Art Release:
October 2026 | v5.1 Final Master Spec

1. Problem Statement: Why Monolithic Chains and Token Bridges Fail AI & Institutional Finance

The global digital economy is undergoing two concurrent structural transformations: Artificial Intelligence agents are evolving from passive chat interfaces into autonomous economic actors capable of licensing data and settling B2B supply-chain obligations, while traditional financial instruments—U.S. Treasuries, commercial real estate, private credit, and commodities—are migrating on-chain via **Real-World Asset (RWA) Tokenization**. Yet existing blockchain architectures fail to unify these workflows due to four foundational engineering bottlenecks:

1. **The Monolithic Engineering Trilemma:** No single blockchain can simultaneously optimize for Proof-of-Work monetary immutability (Bitcoin: ~60-minute settlement), Turing-complete modular smart-account and RWA compliance (Ethereum: 12-second blocks), sub-second machine micro-payment velocity (Solana: 400ms finality, $0.00025 fees; Sui Mysticeti: ~350ms finality), and wholesale cross-border FX liquidity (XRP Ledger: 3–5s finality).
2. **Catastrophic Vulnerability of Lock-and-Mint Bridges:** Traditional cross-chain bridges lock native collateral inside centralized multi-signature vaults or monolithic smart contracts to mint synthetic "wrapped" tokens, creating multi-billion-dollar honey-pots that have accounted for the largest exploits in decentralized finance history.
3. **Unconstrained AI Execution Risk & Dead Working Capital:** Handing a standard private key (`EOA`) to a probabilistic Large Language Model (LLM) exposes corporate treasuries to prompt injection and unauthorized wallet drainage. Simultaneously, holding AI agent operating balances in uninvested stablecoins forfeits the 4.5%–8.0% institutional yield available in compliant tokenized RWAs (ERC-3643 / ERC-4626).
4. **Surveillance Exposure & Quantum Vulnerability:** Transparent blockchains broadcast every corporate RWA balance, supplier invoice, and treasury rebalance to competitors in plain text, while legacy elliptic-curve signatures (`secp256k1` and `Ed25519`) face long-term exposure to Shor's algorithm on cryptographically relevant quantum computers (CRQCs).

2. The "Zero-Khichdi" Design: Lean Intent Core vs. Pluggable Target Adapters

A critical engineering discipline of the IRA Protocol is avoiding the complexity trap of hard-wiring eight disparate blockchains together on Day 1. Instead, IRA adopts the proven systems-engineering pattern of the **Microkernel & Pluggable Device Driver Bus ("USB-C for Blockchains")**. **The IRA Core never merges or hard-codes external blockchains together.** The core protocol is a compact, immutable coordination layer (~1,500 lines of Solidity and TypeScript) that standardizes only three primitives: (a) modular smart-account guardrails (ERC-7579), (b) declarative cross-chain intents (ERC-7683), and (c) verifiable settlement receipts (Chainlink CCIP 2.0). Underlying networks plug into isolated adapter slots—allowing the **Phase-2 Golden Slice MVP** to launch with just Ethereum, XRPL, Solana, and one RWA vault before expanding into the full eight-adapter target topology:

2.1 Anti-Censorship & Zero Single-Parent Dependency (Why No Chain Can Block IRA)

- **Cryptographic Indistinguishability at the Base Layer:** When an IRA solver or ICP Chain Fusion threshold canister settles an intent on Solana, Sui, XRPL, or Bitcoin, it broadcasts standard native `Ed25519`, `secp256k1`, or `BIP-340 Schnorr` transactions. To base-layer validators on Solana or Sui, an IRA settlement is mathematically indistinguishable from any ordinary retail wallet transfer—making selective protocol-level censorship impractical without halting general retail traffic on their own public network.
- **Sub-100ms Algorithmic Circuit-Breaker Failover:** Every adapter slot specifies a hot-standby sister network inside the ERC-7683 `orderData` preference array (e.g., `[Solana_x402, Sui_Mysticeti, Arbitrum_x402]`). If Solana experiences congestion or RPC throttling, competing solvers automatically fulfill the intent on Sui Mysticeti in ~350 milliseconds without requiring the AI agent to re-sign.

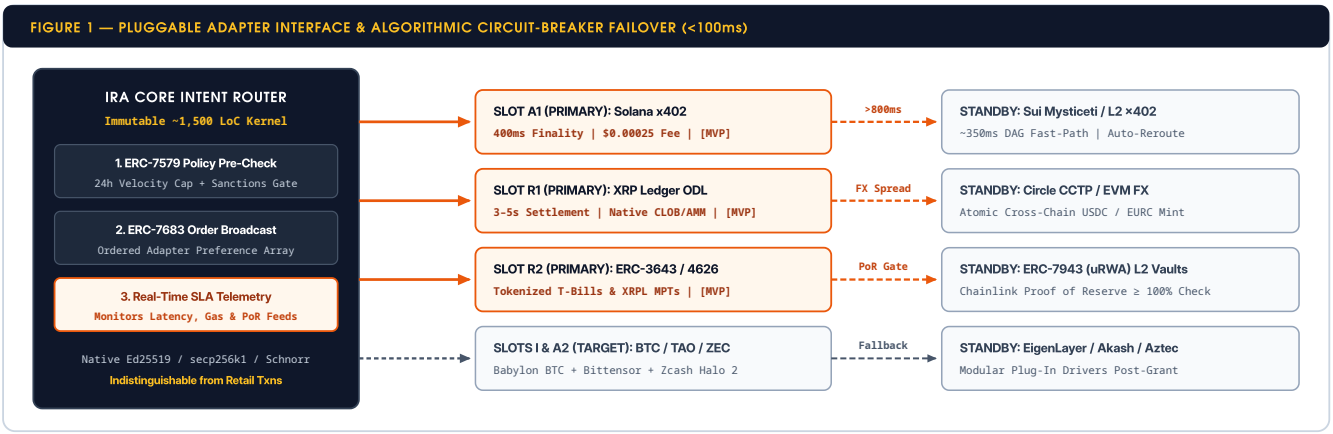


Figure 1: The IRA Microkernel & Pluggable Adapter Bus. Highlighted rows denote the active Phase-2 Golden Slice MVP adapters (Solana x402, XRPL ODL, and ERC-3643/4626 RWA Vaults), alongside automatic sub-100ms failover triggers and post-grant Target Expansion adapters.

Adapter Slot & Scope	Primary Network Driver	Hot-Swap Standby Driver	Deterministic Router Failover Trigger
Slot A1 [MVP]: Fast Micro-Pay	Solana (x402 400ms)	Sui (Mysticeti ~350ms) or Arbitrum/Base x402	Automatic sub-100ms solver rerouting if primary chain slot confirmation latency exceeds 800ms.
Slot R1 [MVP]: Logic & FX	Ethereum & XRP Ledger (ODL / Native AMM)	Arbitrum / Base L2s & Circle CCTP FX Corridors	Ethereum L1 gas spike above threshold or tighter wholesale cross-currency FX spread on standby rail.
Slot R2 [MVP]: RWA Vaults	Ethereum ERC-3643 / ERC-4626 & XRPL MPTs	ERC-7943 (uRWA) on EVM L2s	Continuous Chainlink Proof of Reserve (PoR) verification requiring ≥100% off-chain collateral backing.
Slots I & A2 [Target]: Vault, AI & ZK	Bitcoin (Babylon), ICP, Bittensor & Zcash (Halo 2)	Ethereum EigenLayer Vaults, Akash GPU & Aztec ZK	BTC fee spike, dual-quorum threshold rotation, or Bittensor Yuma consensus confidence score drop.

3. Pillar I Target Adapters: Immutable Reserve (Bitcoin), Sovereign Hosting (ICP) & AI (Bittensor)

In the full Target Architecture, long-term institutional reserves and solver performance bonds are secured without synthetic wrapping by integrating the Babylon Bitcoin Staking Protocol. Native Bitcoin is locked directly on the Bitcoin base chain inside self-custodial Taproot (**`P2TR`**) script trees encoding a timelocked self-custody unbonding branch and a slashable equivocation branch governed by Extractable One-Time Signatures (EOTS). Under EOTS cryptography, signing a single valid state attestation preserves key secrecy; however, if a bonded solver or finality provider double-signs two conflicting settlement roots at the same block height, the reuse of the underlying Schnorr cryptographic nonce mathematically exposes their private key, allowing the protocol to automatically slash the malicious actor's Bitcoin bond while honest depositors retain 100% self-custody.

To eliminate centralized cloud dependencies (AWS/GCP) as the protocol scales, persistent agent coordination loops deploy inside tamper-proof Internet Computer (ICP) canisters utilizing Chain Fusion threshold cryptography (**`secp256k1`** ECDSA, **`BIP-340`** Schnorr, and **`Ed25519`**) to sign native transactions across Bitcoin, Ethereum, Solana, Sui, and XRPL without any single node reconstructing the private key. Market intelligence and solver route-optimization forecasts are sourced from decentralized Bittensor (TAO) subnets validated via stake-weighted Yuma Consensus.

4. Pillar R Adapters: Modular Guardrails (Ethereum), Wholesale FX (XRPL) & RWA Tokenization

4.1 Autonomous Agent Identity (ERC-6551), Guardrails (ERC-7579) & Wholesale FX (XRPL)

In the Phase-2 Golden Slice MVP, every AI agent or corporate sub-treasury is provisioned as a deterministic ERC-6551 Token-Bound Account on Ethereum (**`CREATE2`** address bound to an institutional credential NFT) governed by ERC-7579

Minimal Modular Smart Account hooks and single-chain ERC-8001 Agent Coordination Framework co-signatures (which IRA extends to cross-chain workflows via ERC-7683). Installed ERC-7579 Enforcement Hooks (`IHook`) enforce a mandatory *24-Hour Rolling Spending Cap*, a *Sanctions & Counterparty Whitelist*, and a *Minimum Solver Output Bound* at the EVM bytecode level—making it computationally infeasible for a prompt-injected or hallucinating LLM to drain treasury funds without an authorized cryptographic co-signature. For wholesale B2B cross-border FX clearing across regional stablecoins and fiat corridors, the IRA router utilizes the XRP Ledger (XRPL), combining 3–5 second Federated Consensus with native Central Limit Order Books (CLOB) and XLS-30d Automated Market Makers (On-Demand Liquidity).

4.2 Real-World Asset (RWA) Tokenization & Productive AI Agent Treasuries

A core commercial differentiator of the IRA Protocol is integrating **Real-World Asset (RWA) Tokenization** directly into the AI agent and corporate treasury lifecycle. Institutional assets including U.S. Treasury bills, commercial real estate, gold bullion, and private credit are increasingly tokenized via Ethereum's ERC-3643 (T-REX compliant security token standard, with over $28B tokenized), ERC-7943 (Universal RWA interface), ERC-4626 (Tokenized Yield-Bearing Vaults), and the XRP Ledger's native Multi-Purpose Tokens (MPTs).

Rather than building an isolated RWA issuance platform, IRA plugs tokenized RWAs directly into its existing intent bus across three high-value workflows:

- **1. Zero-Idle-Cash "Productive AI Wallets":** Instead of leaving an AI agent's $50,000 compute budget in 0%-yield cash, the agent's ERC-6551 wallet holds tokenized U.S. Treasury shares (ERC-4626) or XRPL RWA Multi-Purpose Tokens earning 4.5%–5.2% APY continuously.
- **2. Just-in-Time RWA-to-x402 Atomic Solver Swaps:** Whenever the AI agent receives an `HTTP 402` invoice on Solana or an FX invoice on XRPL, it signs an ERC-7683 intent pledging the exact fractional RWA shares required. The IRA `OriginSettler` contract queries Chainlink Proof of Reserve (PoR) to verify ≥100% off-chain custodian backing, and the winning solver immediately fronts native USDC on Solana (400ms) or XRP/RLUSD on XRPL (3–5s).
- **3. Compliant Cross-Chain RWA Settlement with ZK Privacy:** When institutions trade tokenized real estate, bonds, or credit across Ethereum and XRPL, ERC-3643 enforces on-chain KYC whitelists while Zcash Halo 2 viewing keys shield commercial trade sizes from public competitors.

By treating tokenized real-world assets (ERC-3643, ERC-7943, ERC-4626, and XRPL MPTs) as first-class collateral inside the ERC-7683 intent bus, the IRA Protocol transforms static institutional balance sheets into liquid, yield-generating machine capital without manual redemptions.

RWA Collateral Class	On-Chain Token Standard	Verification Oracle & Instant Liquidity Settlement Path
Tokenized U.S. Treasuries & Money Market Funds	Ethereum ERC-3643 / ERC-4626 & XRPL MPT	Verified via Chainlink Proof of Reserve (PoR); fractional shares swapped atomically via ERC-7683 solvers into 400ms Solana USDC (x402).
Tokenized Short-Term Corporate Credit & Invoices	Ethereum ERC-7943 (uRWA) & ERC-7540	Enforces ERC-7579 counterparty KYC hooks; settles cross-border B2B supplier invoices in 3–5 seconds via XRPL ODL.
Tokenized Commodities, Gold & Real Estate	Ethereum ERC-3643 & XRPL Native RWA Tokens	Custodian vault attestations verified on-chain while commercial trade sizes are shielded via Zcash Halo 2 viewing keys.

5. Pillar A Adapters: Sub-Second Velocity (Solana & Sui) & Audited ZK Privacy (Zcash)

To power real-time machine-to-machine API commerce and protect sensitive enterprise data, Pillar A combines high-throughput execution rails with recursive zero-knowledge cryptography:

- **Sub-Second x402 Micro-Payments (Solana & Sui Mysticeti):** Reported 2026 ecosystem telemetry indicates that Solana processes approximately 76% of live x402 AI agent transactions, supported by 400ms slot finality, $0.00025 median fees, and 6,284+ TPS reported benchmark throughput. Paired as an automatic failover twin, Sui's Mysticeti uncertified DAG consensus achieves ~250–500ms finality for owned-object state transitions and has demonstrated 100,000+ TPS in controlled multi-node benchmarks.

- **Audited Zero-Knowledge Commercial Privacy (Zcash Orchard & Halo 2):** Using Zcash's Orchard shielded pool and trustless **Halo 2 recursive zk-SNARKs** over Pasta cycle curves, sender, recipient, amount, and 512-byte encrypted RWA/invoice memos remain confidential on-chain, while **Hierarchical Viewing Keys (IVK/FVK)** provide full mathematical auditability to authorized regulators and compliance officers.

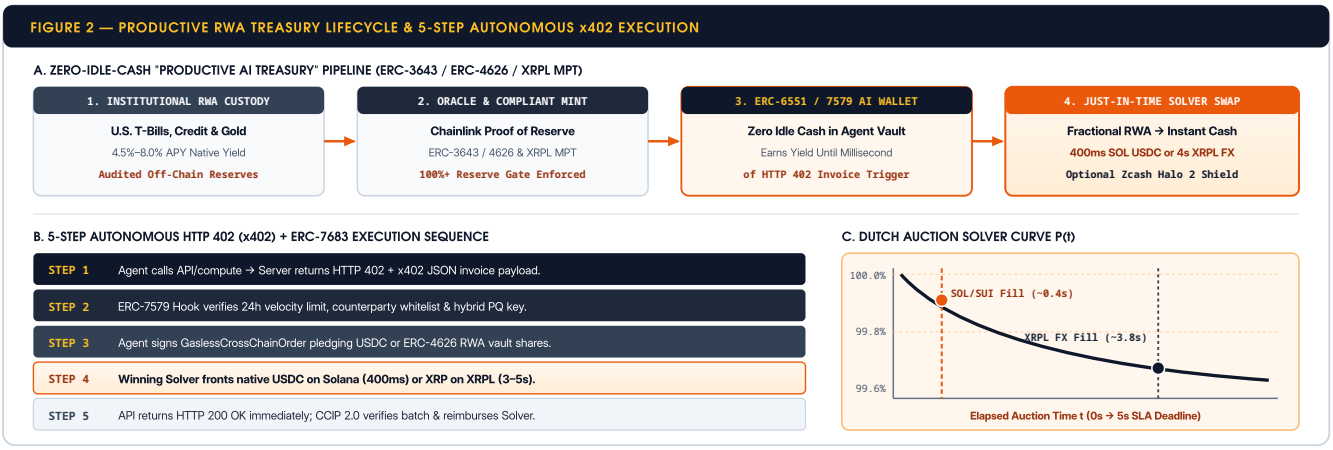


Figure 2: (A) The 4-stage IRA Productive RWA Treasury pipeline. (B) The 5-step autonomous HTTP 402 (x402) + ERC-7683 execution sequence. (C) Dutch auction solver price-discovery curve $P(t)$ driving optimal execution across fast-path (Solana/Sui) and wholesale FX (XRPL) rails.

6. Interoperability Mechanics: ERC-7683 Dutch Auctions, x402 Flow & CCIP 2.0 Security

The IRA Protocol replaces vulnerable token bridges with ERC-7683 Cross-Chain Intents, the HTTP 402 (x402) standard, and Dutch auction solver competition. When an AI agent receives an **`HTTP 402 Payment Required`** challenge, it signs an off-chain ERC-7683 **`GaslessCrossChainOrder`** pledging stablecoins or ERC-4626 RWA vault shares on Ethereum. Following formal Dutch auction mechanics, the solver fulfillment rate decays over $t \in [t_{start}, t_{deadline}]$ according to $P(t) = P_{initial} - (P_{initial} - P_{reserve}) \cdot ((t - t_{start}) / (t_{deadline} - t_{start}))^{\alpha}$ (Figure 2C). Competing solvers race to fill the order using their own native inventory on Solana (400ms), Sui (~350ms), XRPL (3–5s), or Zcash—absorbing all bridging latency and finality risk. Batched settlement receipts are verified via Chainlink CCIP 2.0 and its independent Risk Management Network before releasing net reimbursements from the Ethereum escrow contract.

7. Post-Quantum Cryptographic Agility: Near-Term Hybrid Auth & Emerging EIP-8051 / EIP-8355 Path

While no coordination layer alters base-layer Bitcoin or Ethereum consensus prior to their respective hard forks, IRA builds cryptographic agility directly into **smart-account authorization, RWA vault custody, intent signing, and oracle attestations** using the U.S. NIST-standardized **Module-Lattice-Based Digital Signature Algorithm (ML-DSA / FIPS 204)** over the polynomial ring $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$ $(q = 8{,}380{,}417)$.

Because verifying an ML-DSA signature in pure Solidity currently consumes ~650,000 to 4.8 million gas and proposed EVM precompiles (EIP-8051 and companion proposal EIP-8355) remain in **Draft** status awaiting future hard-fork inclusion, the IRA Protocol implements a pragmatic two-stage post-quantum architecture:

- **Near-Term Production (Hybrid ECDSA + Off-Chain / L2 Lattice Verification):** In the immediate MVP and early mainnet releases, IRA ERC-7579 validator modules enforce hybrid composite signatures $\Sigma_{hybrid} = (\sigma_{ECDSA}, \sigma_{ML\text{-}DSA})$ where the lattice signature is verified off-chain by Chainlink CCIP verifiers / ICP canisters or on low-cost Rollup L2s (~25,000 gas equivalent) prior to origin escrow release.
- **Long-Term Precompile Path (Proposed EIP-8051 / EIP-8355):** As Ethereum adopts native ML-DSA precompiles (such as the Keccak-optimized **`VERIFY_MLDSA_ETH`** specification targeting ~4,500 gas in Figure 3B), IRA's modular ERC-7579 validator upgrades seamlessly in place without migrating user accounts or RWA vaults.

8. Grant-First Execution Roadmap & Long-Term B2B Infrastructure Economics

Launching a speculative token prematurely invites regulatory friction and disqualifies infrastructure projects from non-commercial foundation grants such as the Ethereum Foundation ESP. Equally important, realistic infrastructure commercialization follows a **Grant-First, SaaS-Second, Enterprise-Third** trajectory: early capital (Months 3–6) is secured through non-dilutive ecosystem grants ($10,000–$200,000) to fund security audits and engineering, which then unlocks five recurring commercial revenue streams in Phase 4—(1) *Intent Solver Routing Fees* (0.5–2.0 bps), (2) *RWA-to-x402 Liquidity Conversion Spreads* (2–5 bps), (3) *Developer SDK Subscriptions* ($49–$499/mo post-audit), (4) *Institutional ZK & CCIP Verifier SLAs* ($2,500–$15,000/mo), and (5) *Dutch Auction Solver Rebates*. Crucially, while the reference core is open-source, the protocol's long-term **Competitive Moat** rests on three compounding advantages that a code fork cannot replicate: (a) *Prior-Art & Canonical Standard Stewardship* (establishing the neutral I-R-A adapter interface adopted by wallets and grant foundations, analogous to WalletConnect or OAuth), (b) *Two-Sided Solver & RWA Issuer Liquidity Network Effects* (where deeper solver competition lowers execution spreads, attracting more ERC-3643 / XRPL MPT issuers), and (c) *Accumulated Compliance & Telemetry Registry State* (Chainlink PoR bindings, audited ERC-7579 policy templates, and solver reputation history):

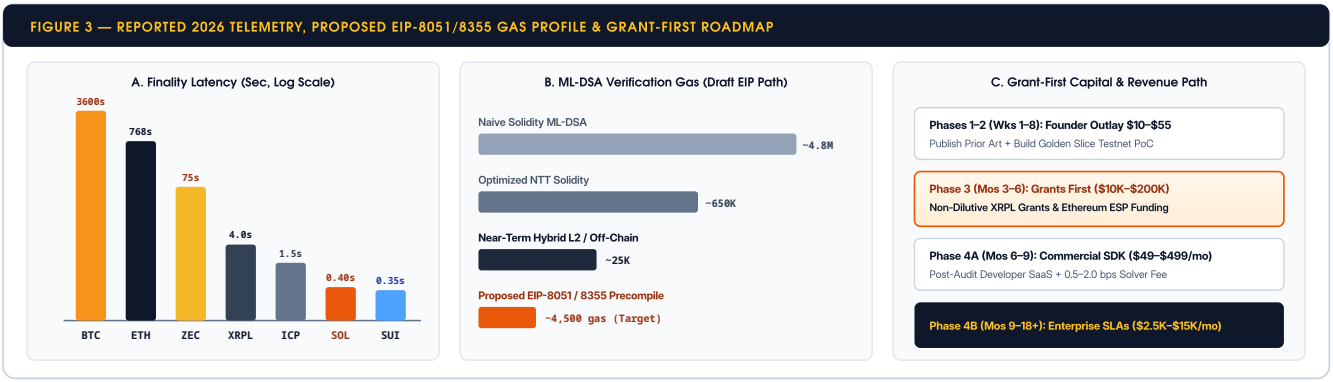


Figure 3: (A) Reported 2026 finality latency across pluggable network adapters. (B) Post-quantum verification gas profile comparing naive Solidity, near-term hybrid L2/off-chain verification, and the proposed EIP-8051 / EIP-8355 precompile target (~4,500 gas). (C) Realistic grant-first capital progression from $10–$55 founder outlay to non-dilutive grants and post-audit commercial revenue.

Execution Phase	Timeline & Cost	Technical Scope (MVP vs. Target Expansion)	Realistic Funding & Revenue Source
Phase 1: Prior Art & Brand Launch	Weeks 1–2 **$10 – $35**	Commit this whitepaper to a timestamped GitHub repo; publish publicly on Medium & Substack (mandatory for legal prior-art priority); launch portal on custom domain (Namecheap).	Establish verifiable public prior art; open developer waitlist ($0 revenue stage).
Phase 2: "Golden Slice" Testnet MVP	Weeks 3–8 **$0 – $20**	Build strictly the **3-Chain + 1-RWA MVP**: **`IRAOriginSettler.sol`** (ERC-7683), ERC-7579 Hook & 1 mock ERC-4626 RWA vault on Ethereum Sepolia (Remix IDE), routing x402 agent payments to Solana Devnet & XRPL Testnet.	Working open-source prototype required to qualify for ecosystem grants & hackathons.
Phase 3: Non-Dilutive Grants (Primary Early Capital)	Months 3–6 **$0 – $500**	Submit working Phase-2 Golden Slice MVP + GitHub repos for technical review; add Sui Mysticeti failover driver.	**Primary Early Funding:** XRPL Grants ($10K–$200K for RWA MPT + AI), Ethereum ESP, and Solana grants.
Phase 4: Mainnet Audits & Target Expansion	Months 6–18+ **Grant-Funded**	Use grant capital to fund security audits ($15K–$100K+) and expand into Target Architecture (Babylon BTC, ICP, Bittensor, Zcash Halo 2).	**Commercial Monetization:** Paid Developer SDK ($49–$499/mo), solver routing bps, & bank SLAs ($2.5K–$15K/mo).

9. Defense-in-Depth Threat Mitigation Matrix & Institutional Use Cases

Applying the adversarial verification framework of the Chainlink CCIP Risk Management Network and Satoshi Nakamoto's threat modeling, the IRA Protocol isolates every failure domain so that a failure in any single chain or RWA issuer cannot compromise the broader system:

Adversarial or Systemic Threat	Legacy Single-Chain / Bridge Outcome	IRA Protocol Cryptographic & Architectural Mitigation
1. Execution Chain Outage or Censorship (e.g., Solana or Sui halts or blocks RPCs)	Complete application downtime; trapped user collateral; frozen AI agent operations.	**Sub-100ms Adapter Failover:** Solvers automatically reroute ERC-7683 order fulfillment from Solana to Sui Mysticeti or XRPL using native **Ed25519** signatures indistinguishable from retail traffic.
2. Prompt Injection or Rogue AI Hallucination	Unconstrained private key (**EOA**) drains entire corporate treasury to attacker address.	**Deterministic ERC-7579 Guardrails:** On-chain Velocity Cap Hook and Counterparty Whitelist Hook reject unauthorized transfers at the EVM bytecode level regardless of LLM output.
3. Under-Collateralized or Unverified RWA Token	De-pegged or fraudulent synthetic asset infects cross-chain liquidity pools.	**Chainlink Proof of Reserve (PoR) + ERC-3643 Gate:** Origin escrow contract automatically halts RWA-backed intents if Chainlink PoR reports off-chain custodian reserves below 100%.
4. Malicious Solver Equivocation or Default	Bridge insolvency or unbacked synthetic wrapped tokens on destination chain.	**Zero-Wrap Native Fronting + Babylon Slashing:** Solvers front their own native funds first; double-signing in Target Architecture exposes their Babylon EOTS key and slashes their BTC bond.
5. Competitor Surveillance & Quantum Cryptanalysis	Competitors scrape plain-text RWA holdings; quantum Shor's algorithm forges ECDSA keys.	**Audited Zcash Halo 2 Privacy + Hybrid Lattice Auth:** Zcash Orchard shields trade data (with regulator Viewing Keys), while hybrid ECDSA + ML-DSA protects vaults along the EIP-8051 / EIP-8355 path.

10. References and Canonical Technical Standards Index

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3. Real-World Asset (RWA) Tokenization Standards: **ERC-3643 (T-REX Token for Regulated Exchanges)**, **ERC-7943 (uRWA Universal RWA Interface)**, **ERC-4626 (Tokenized Yield-Bearing Vaults)**, **XRPL Multi-Purpose Tokens (MPTs)**, and **Chainlink Proof of Reserve (PoR)**.
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